Technical Document #329: Natural Language Processing - Comprehensive Overview

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Question answering systems extract answers from a given context. They typically consist of a retriever component and a reader component.

Sentiment analysis determines the emotional tone behind a body of text. It is widely used in social media monitoring and customer feedback analysis.

Named entity recognition (NER) identifies and classifies named entities in text into predefined categories such as person names, organizations, and locations.

Language models predict the probability of a sequence of words. Large language models like GPT and BERT have achieved remarkable performance on various NLP tasks.

Word embeddings represent words as dense vectors in continuous space. Word2Vec, GloVe, and FastText are popular word embedding techniques.

Tokenization is the process of breaking text into individual units called tokens. Subword tokenization methods like BPE handle out-of-vocabulary words gracefully.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

Natural language processing has made significant advances with large language models. These models can understand and generate human-like text with remarkable fluency.

Sustainability and environmental responsibility are becoming key considerations in technology design. Green computing aims to reduce the environmental impact of technology.

The rapid advancement of technology has transformed how organizations approach problem-solving and innovation. Companies must adapt to stay competitive in an ever-evolving landscape.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Key Takeaways:

- Sentiment analysis determines the emotional tone behind a body of text.
- Question answering systems extract answers from a given context.
- Text summarization generates a concise summary of a longer text.